CSC1042 Computer Network II

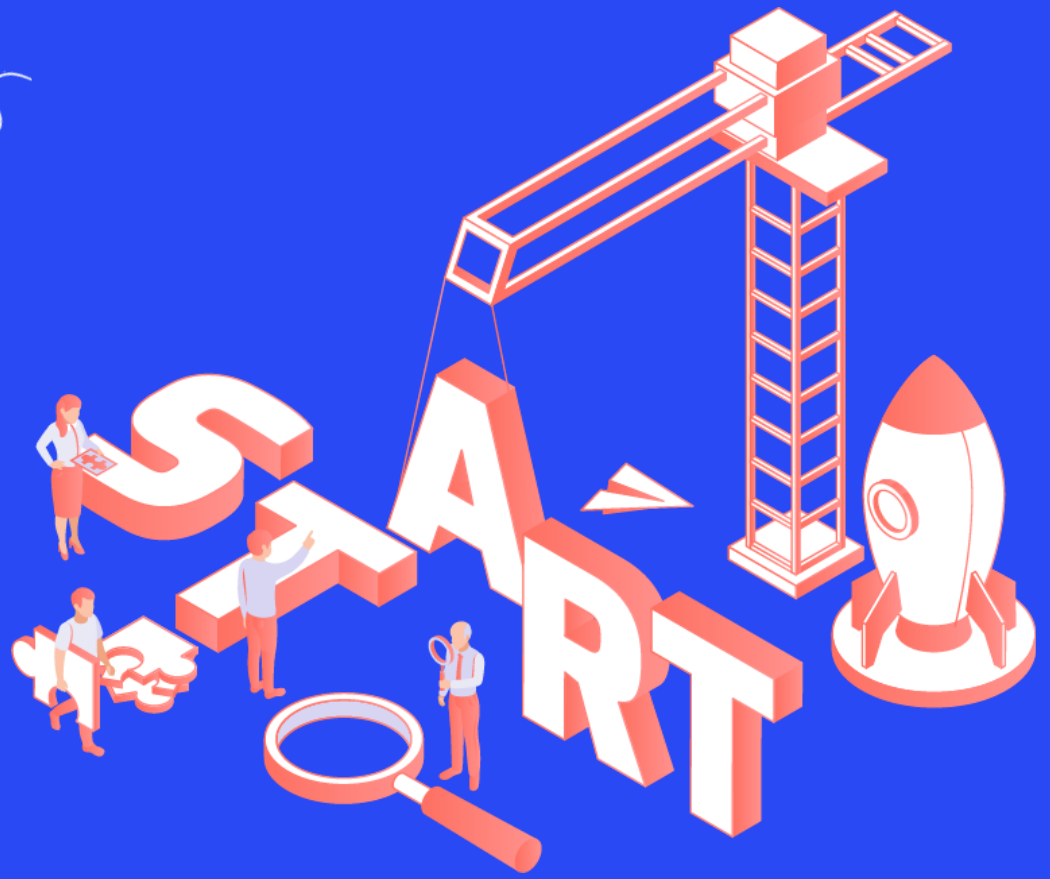
Dijkstra's Algorithm

Dr. Mohammed Amine TOGOU



Learning outcomes

- Understand how Dijkstra's algorithm compute the shortest path between two nodes

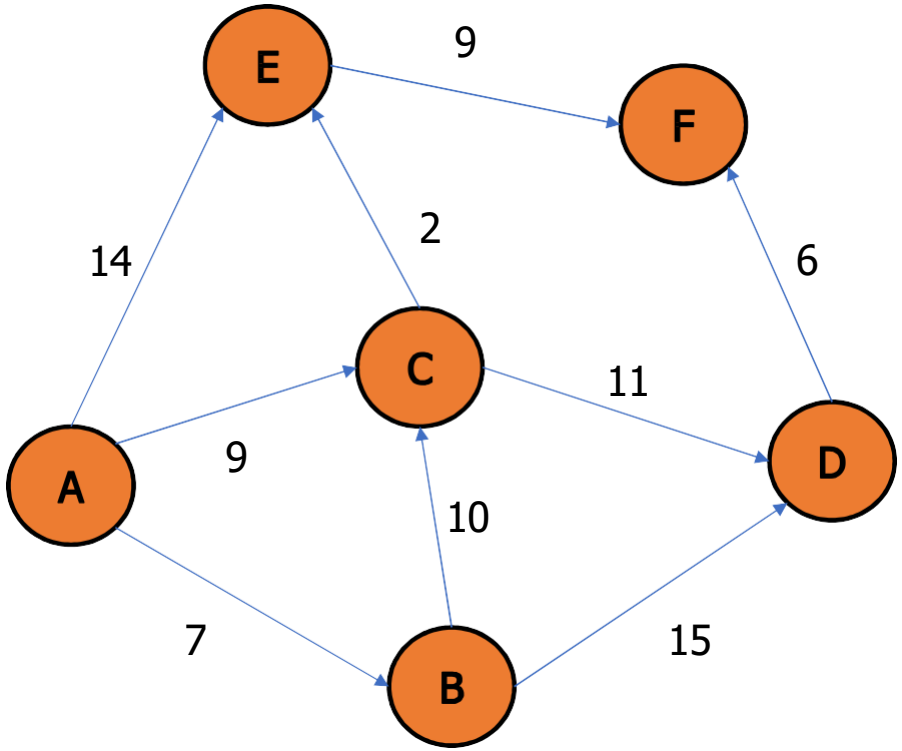


Dijkstra's algorithm

- It allows to find the shortest path between two nodes on a graph
- Input: the graph along with start and finish nodes
  - Edges on the graph are weighted
- Output: the shortest path across the graph between the start and finish nodes

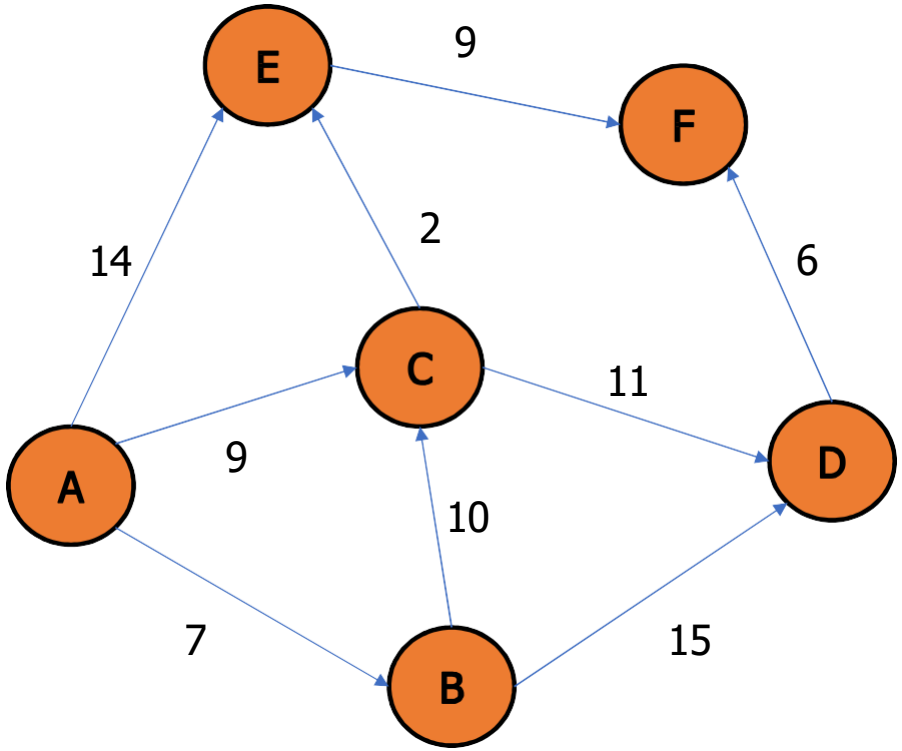
Shortest paths

- Let's look at this graph
- What is the shortest path from node A to all nodes in the graph?



Dijkstra's algorithm

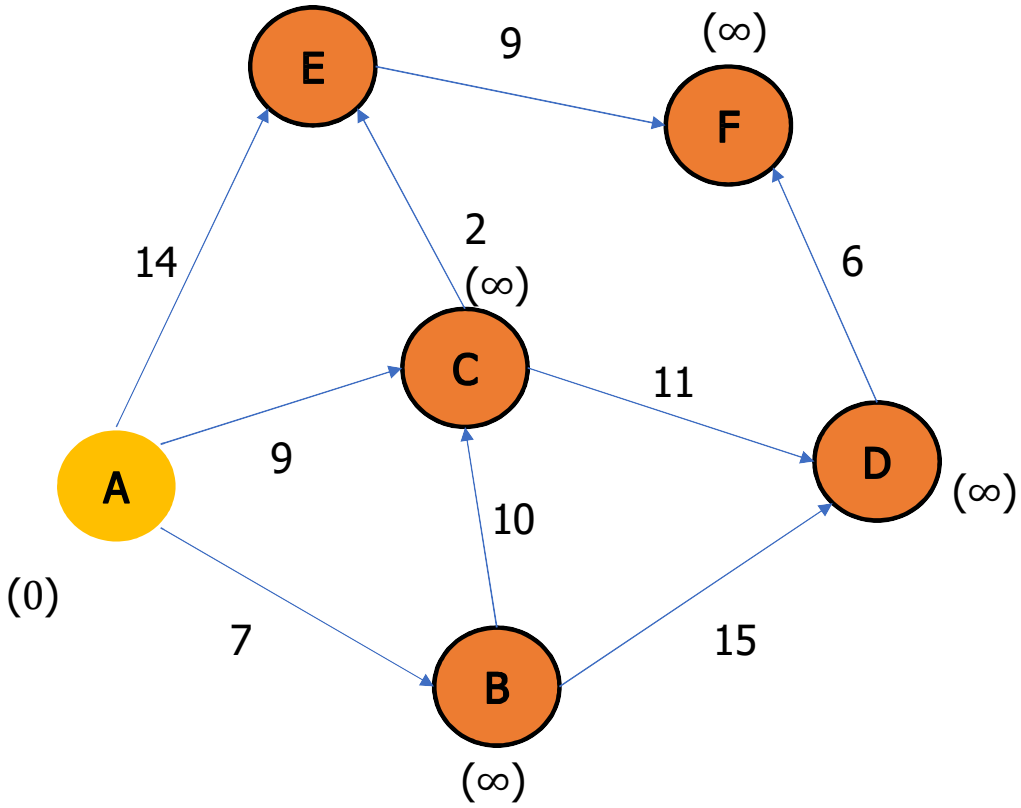
- Step 1: initialise node A with distance 0 and remaining nodes with distance ∞
- Step 2: update distances of A's unvisited neighbours
- Step 3: select the neighbour with the shortest distance and mark it as visited
- Step 4: repeat steps 2 and 3 for the current node



Dijkstra's algorithm

- Step 1: initialise node A with distance 0 and remaining nodes with distance ∞

Node	Shortest Distance	Previous Node
A	0	
B	∞	
C	∞	
D	∞	
E	∞	
F	∞	



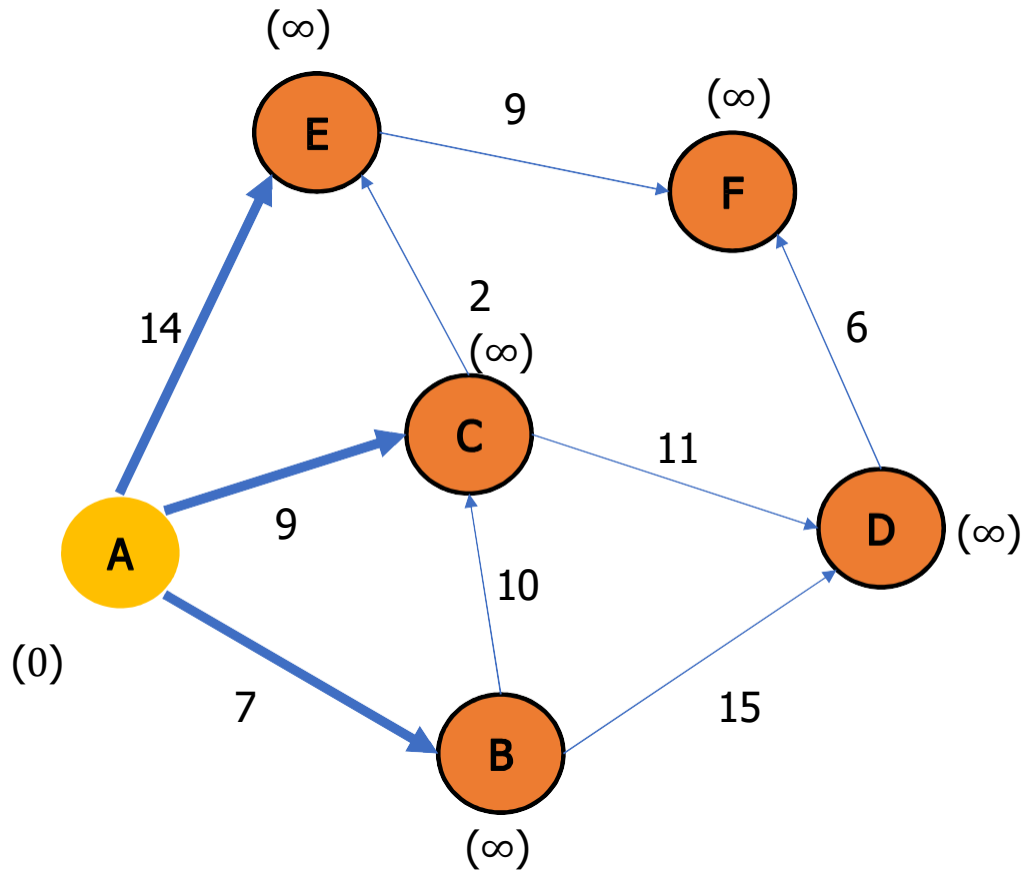
Visited Nodes: []

Unvisited Nodes: [A,B,C,D,E,F]

Dijkstra's algorithm

- Step 2: update distances of A's unvisited neighbours

Node	Shortest Distance	Previous Node
A	0	
B	∞	
C	∞	
D	∞	
E	∞	
F	∞	

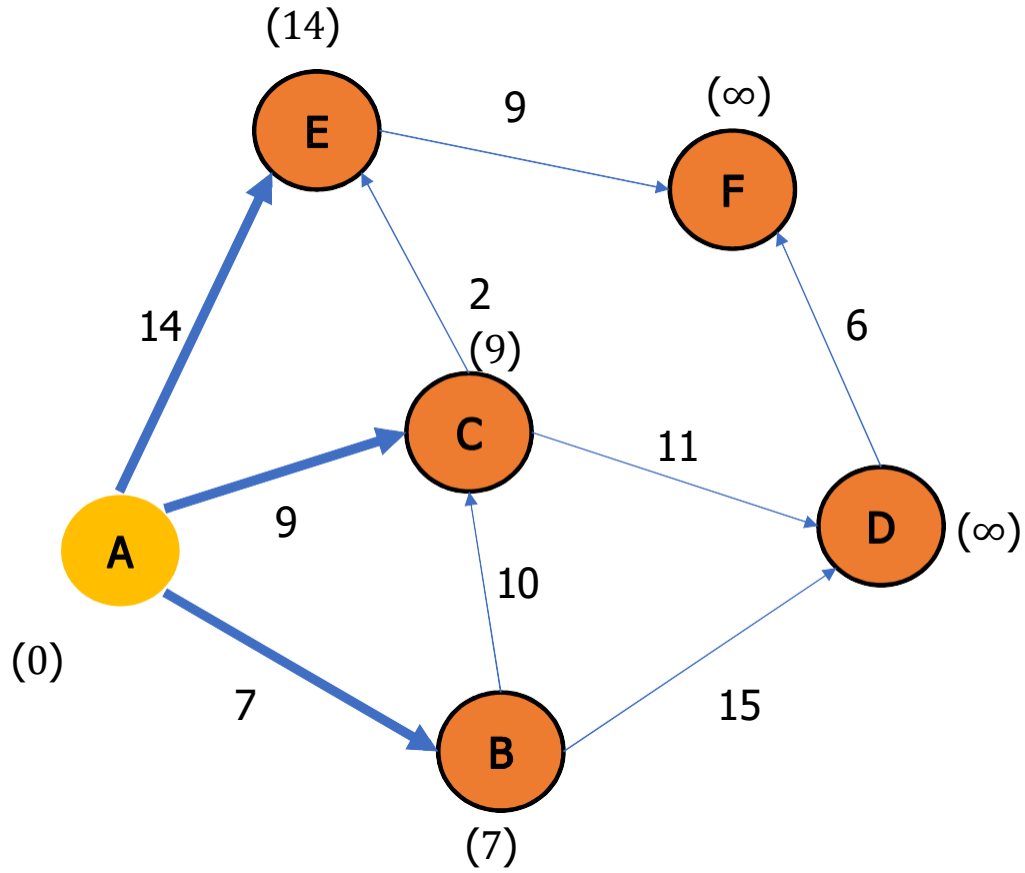


Visited Nodes: [] Unvisited Nodes: [A,B,C,D,E,F]

Dijkstra's algorithm

- Step 2: update distances of A's unvisited neighbours

Node	Shortest Distance	Previous Node
A	0	
B	7	A
C	9	A
D	∞	
E	14	A
F	∞	



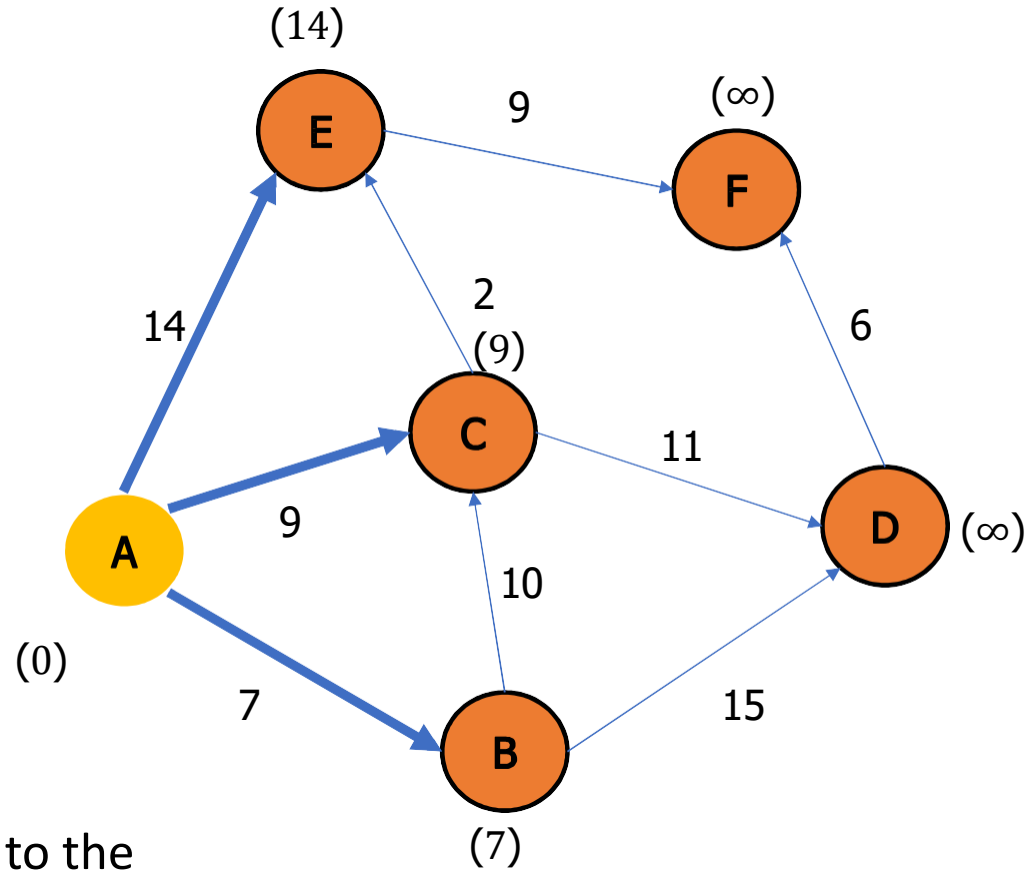
Visited Nodes: []

Unvisited Nodes: [A,B,C,D,E,F]

Dijkstra's algorithm

- Step 2: update distances of A's unvisited neighbours

Node	Shortest Distance	Previous Node
A	0	
B	7	A
C	9	A
D	∞	
E	14	A
F	∞	



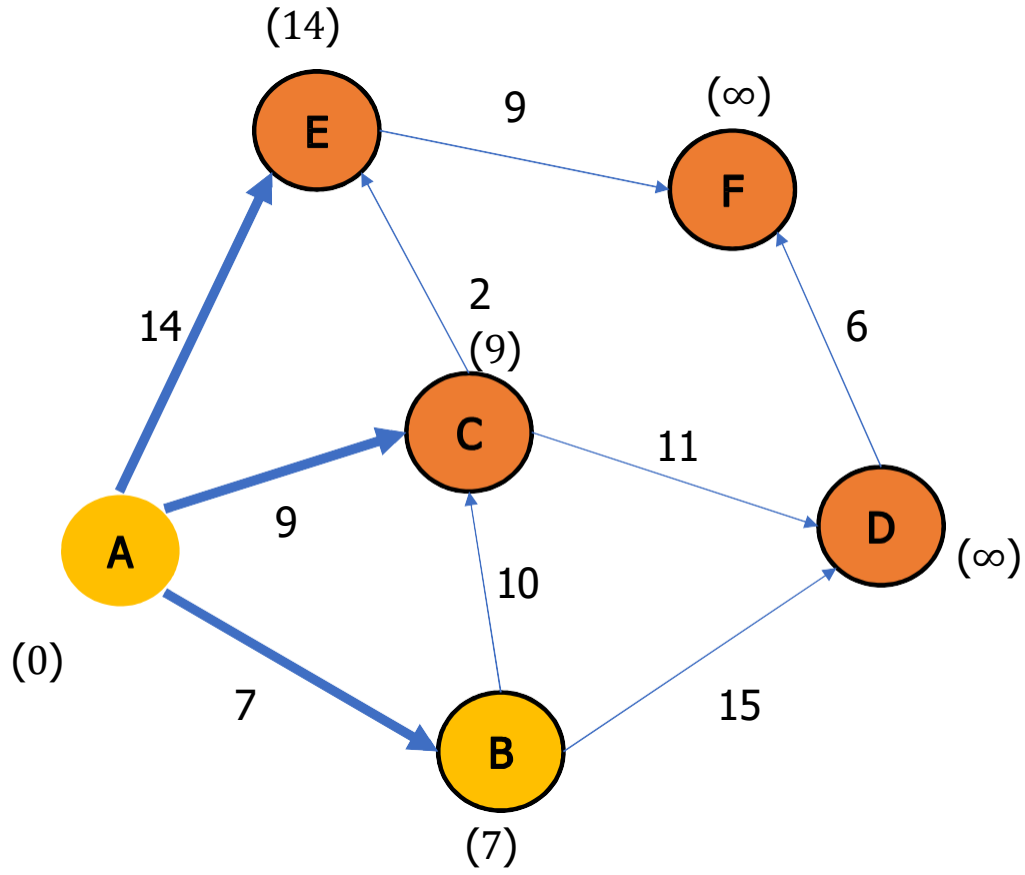
- Once all distances of A's neighbours are updated, we add A to the Visited Nodes list

Visited Nodes: [A] Unvisited Nodes: [B,C,D,E,F]

Dijkstra's algorithm

- Step 3: select the neighbour with the shortest distance

Node	Shortest Distance	Previous Node
A	0	
B	7	A
C	9	A
D	∞	
E	14	A
F	∞	



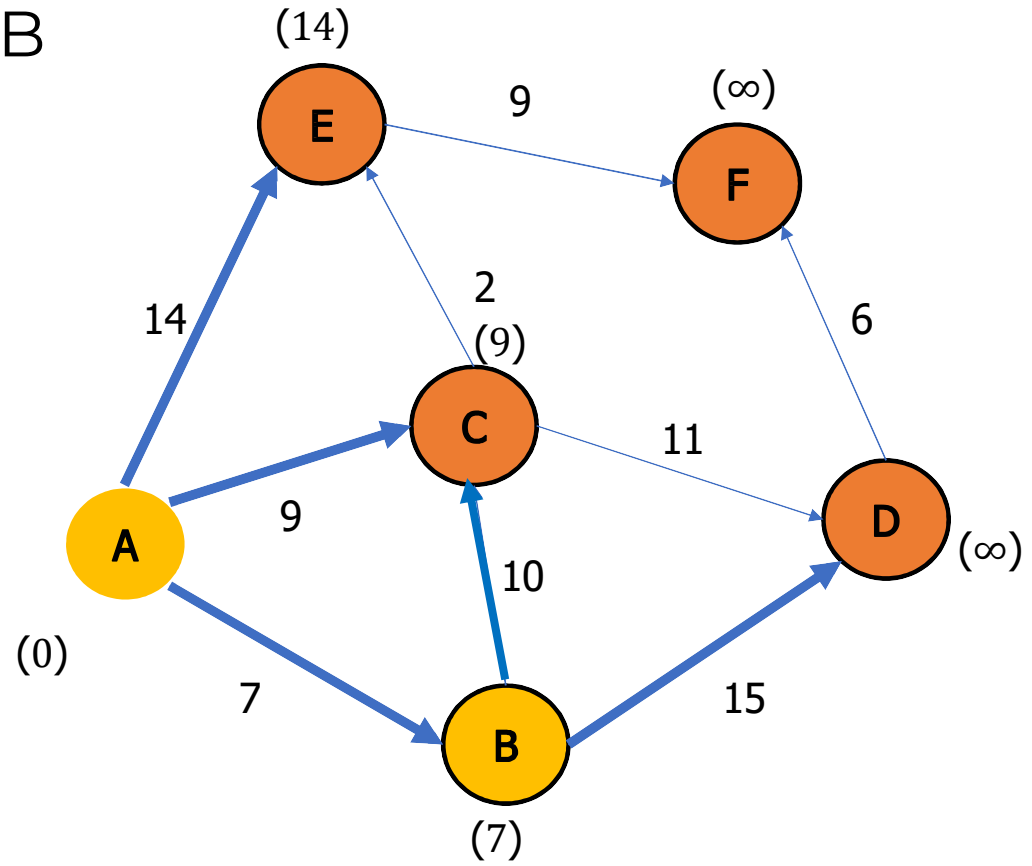
Visited Nodes: [A]

Unvisited Nodes: [B,C,D,E,F]

Dijkstra's algorithm

- Step 4: repeat steps 2 and 3 from node B

Node	Shortest Distance	Previous Node
A	0	
B	7	A
C	9	A
D	∞	
E	14	A
F	∞	



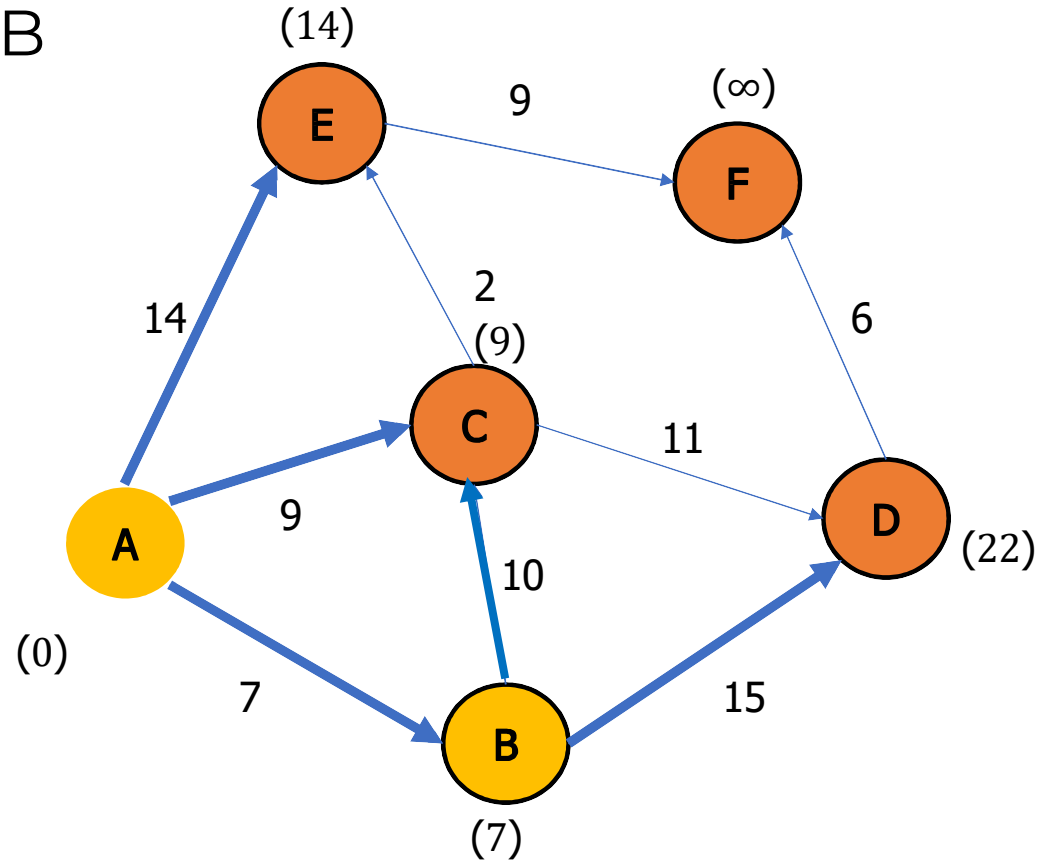
Visited Nodes: [A] Unvisited Nodes: [B,C,D,E,F]

Dijkstra's algorithm

- Step 4: repeat steps 2 and 3 from node B

Node	Shortest Distance	Previous Node
A	0	
B	7	A
C	9	A
D	7+15 = 22	B
E	14	A
F	∞	

C can be reached through B with a cost of 17 > 9, so the table is not updated

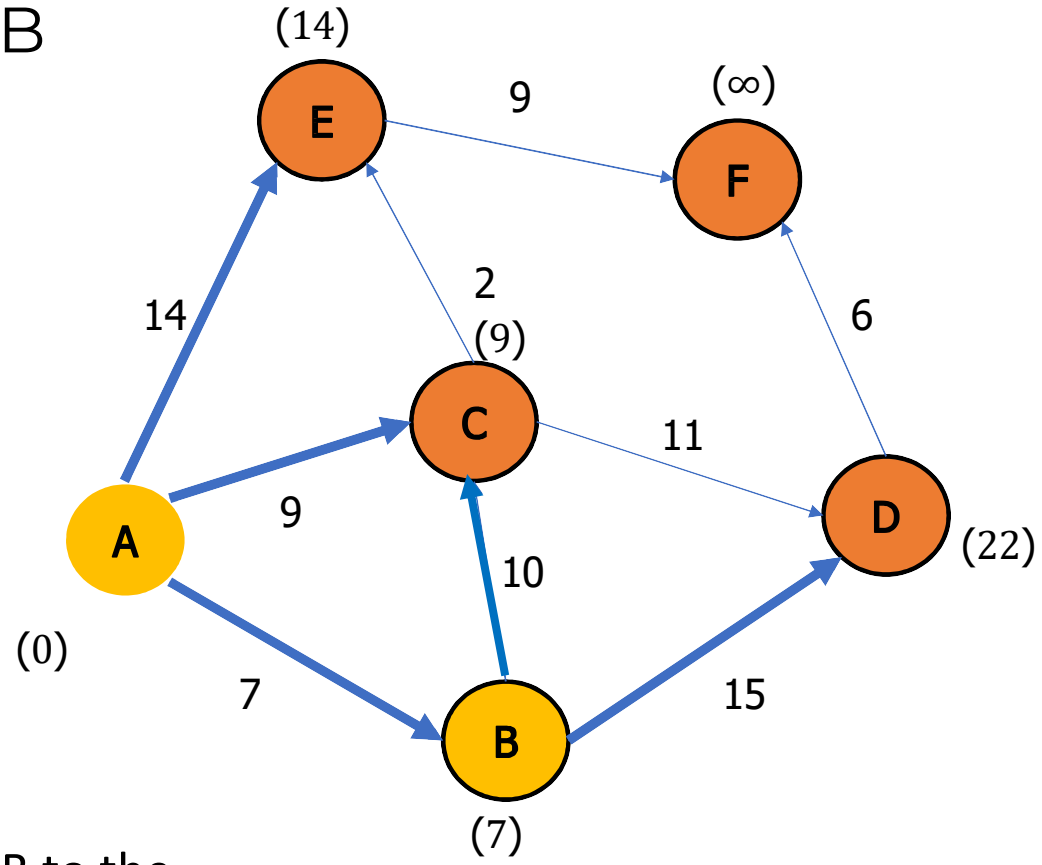


Visited Nodes: [A] Unvisited Nodes: [B,C,D,E,F]

Dijkstra's algorithm

- Step 4: repeat steps 2 and 3 from node B

Node	Shortest Distance	Previous Node
A	0	
B	7	A
C	9	A
D	22	B
E	14	A
F	∞	



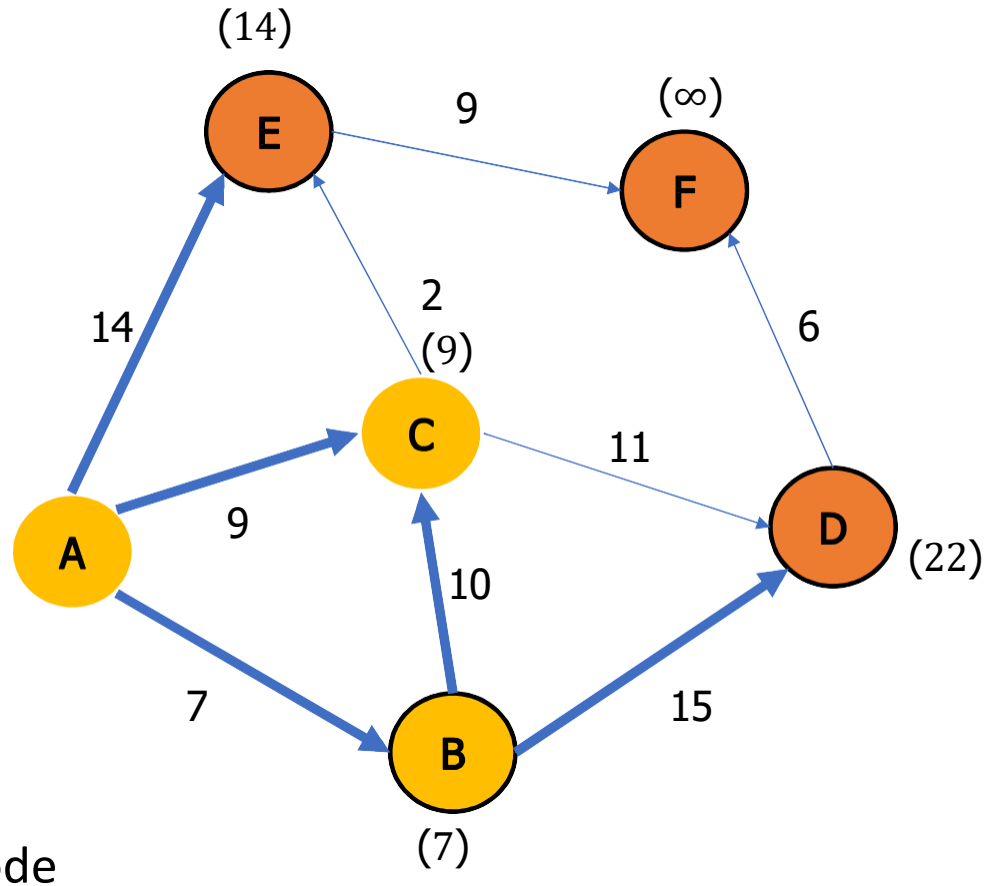
- Once all distances of B's neighbours are updated, we add B to the Visited Nodes list

Visited Nodes: [A,B] Unvisited Nodes: [C,D,E,F]

Dijkstra's algorithm

- Step 4: repeat steps 2 and 3 from node B

Node	Shortest Distance	Previous Node
A	0	
B	7	A
C	9	A
D	22	B
E	14	A
F	∞	



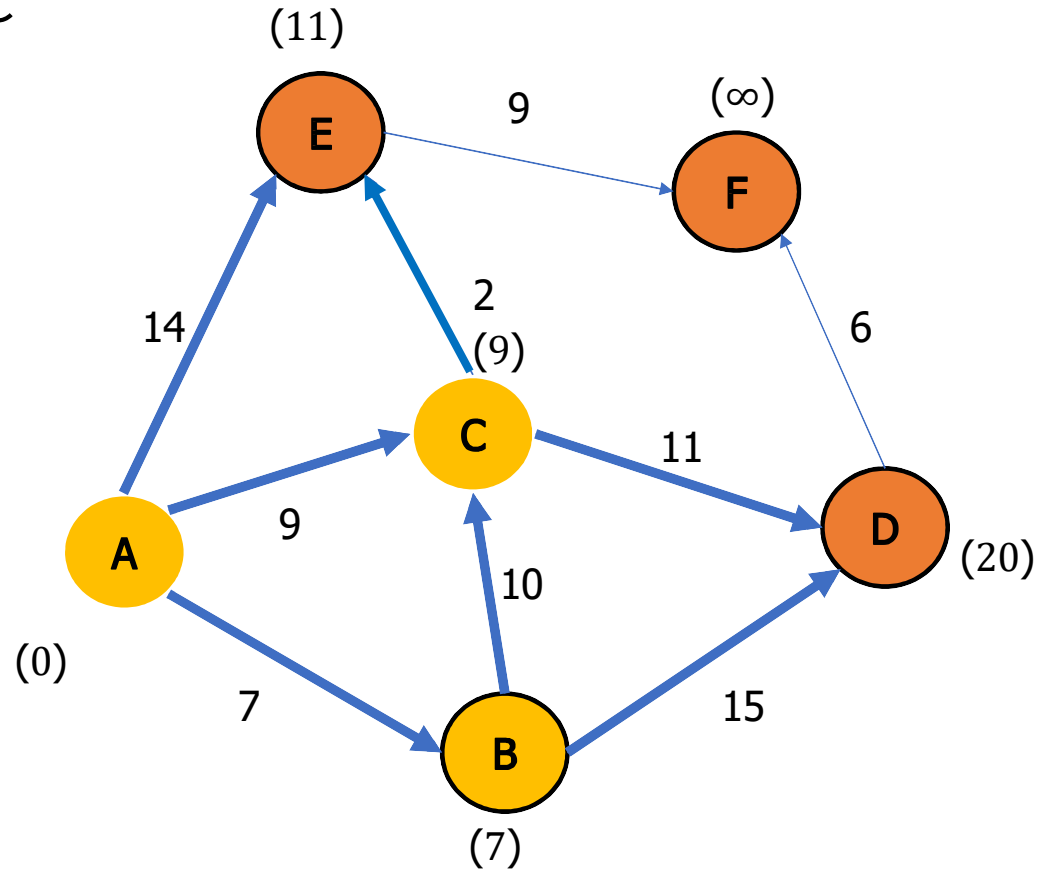
- Since C has the shortest distance, it is selected as the next node

Visited Nodes: [A,B] Unvisited Nodes: [C,D,E,F]

Dijkstra's algorithm

- Step 4: repeat steps 2 and 3 from node C

Node	Shortest Distance	Previous Node
A	0	
B	7	A
C	9	A
D	~~22~~ 20	C
E	~~14~~ 11	C
F	∞	

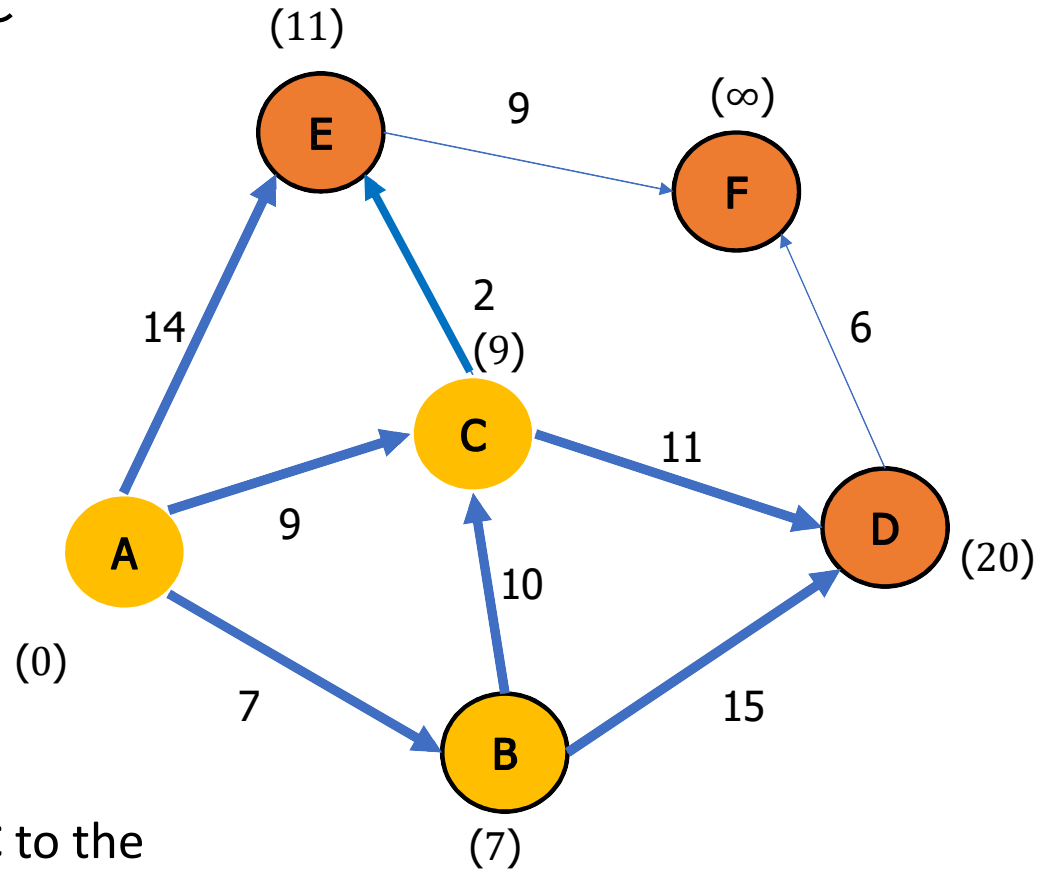


Visited Nodes: [A,B] Unvisited Nodes: [C,D,E,F]

Dijkstra's algorithm

- Step 4: repeat steps 2 and 3 from node C

Node	Shortest Distance	Previous Node
A	0	
B	7	A
C	9	A
D	20	C
E	11	C
F	∞	



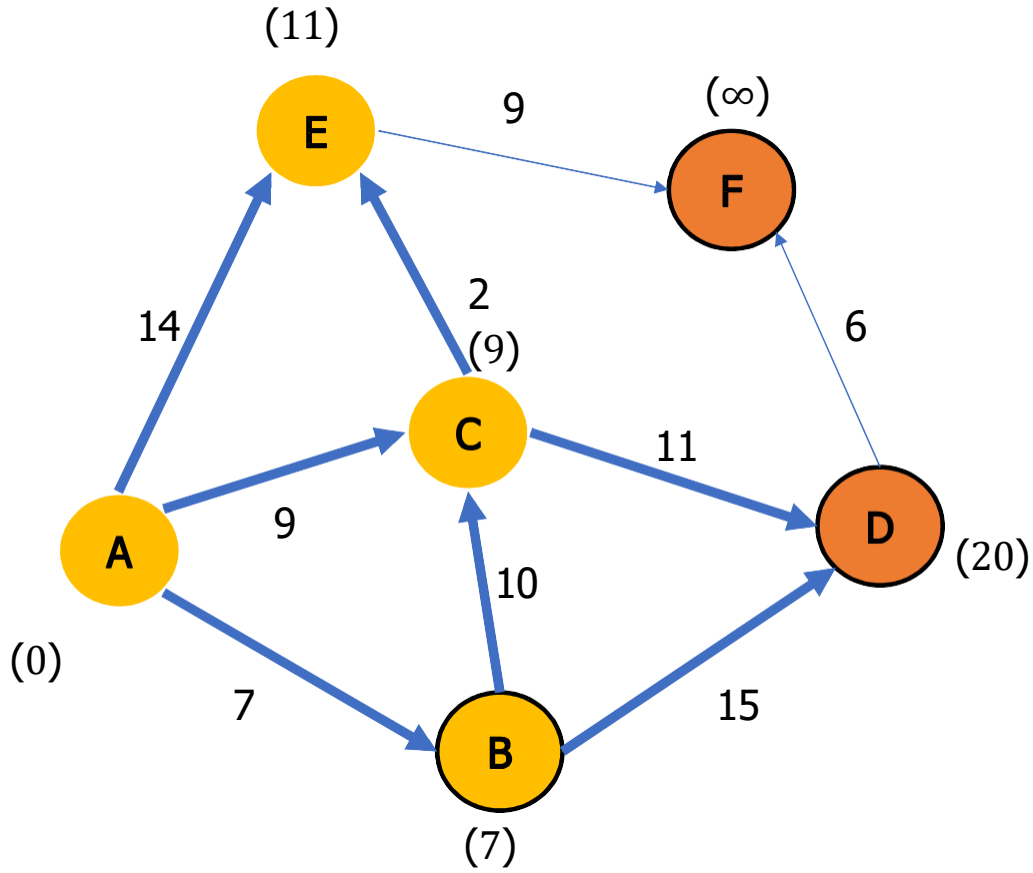
- Once all distances of C's neighbours are updated, we add C to the Visited Nodes list

Visited Nodes: [A,B,C] Unvisited Nodes: [D,E,F]

Dijkstra's algorithm

- Step 4: repeat steps 2 and 3 from node C

Node	Shortest Distance	Previous Node
A	0	
B	7	A
C	9	A
D	20	C
E	11	C
F	∞	



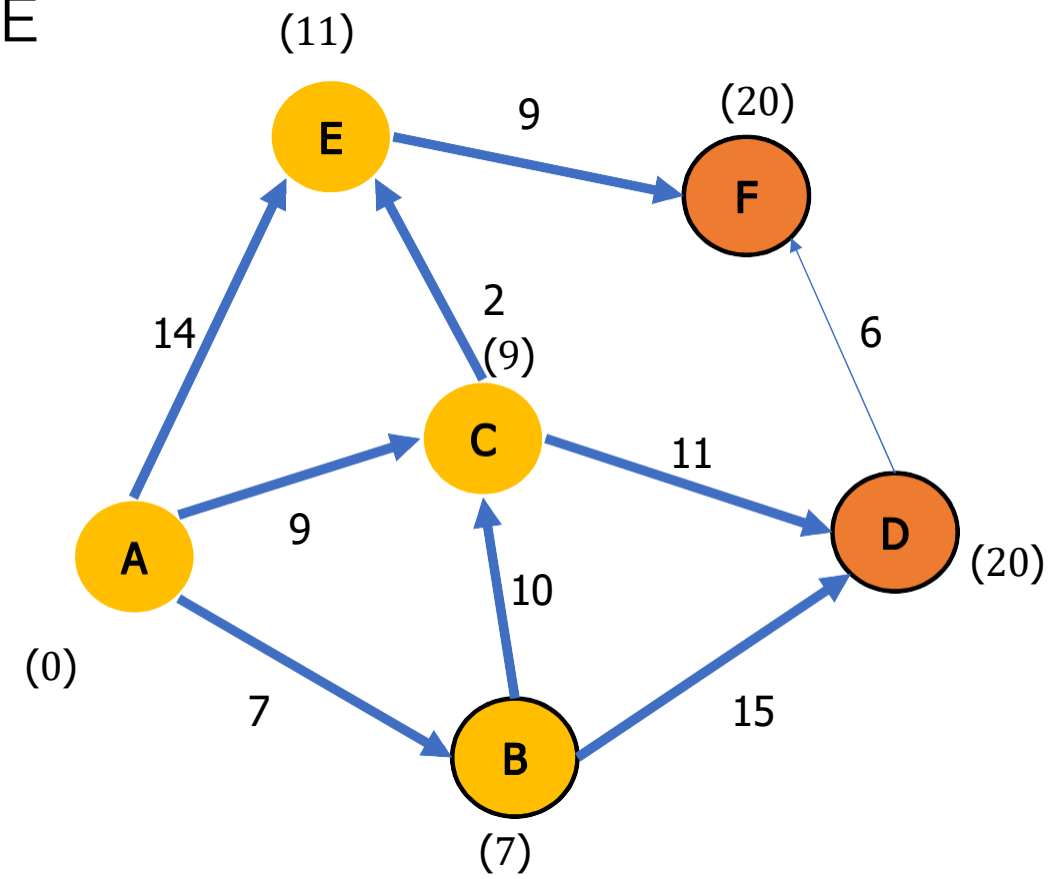
- Since E has the shortest distance, it is selected as the next node

Visited Nodes: [A,B,C] Unvisited Nodes: [D,E,F]

Dijkstra's algorithm

- Step 4: repeat steps 2 and 3 from node E

Node	Shortest Distance	Previous Node
A	0	
B	7	A
C	9	A
D	20	C
E	11	C
F	11+9 = 20	E

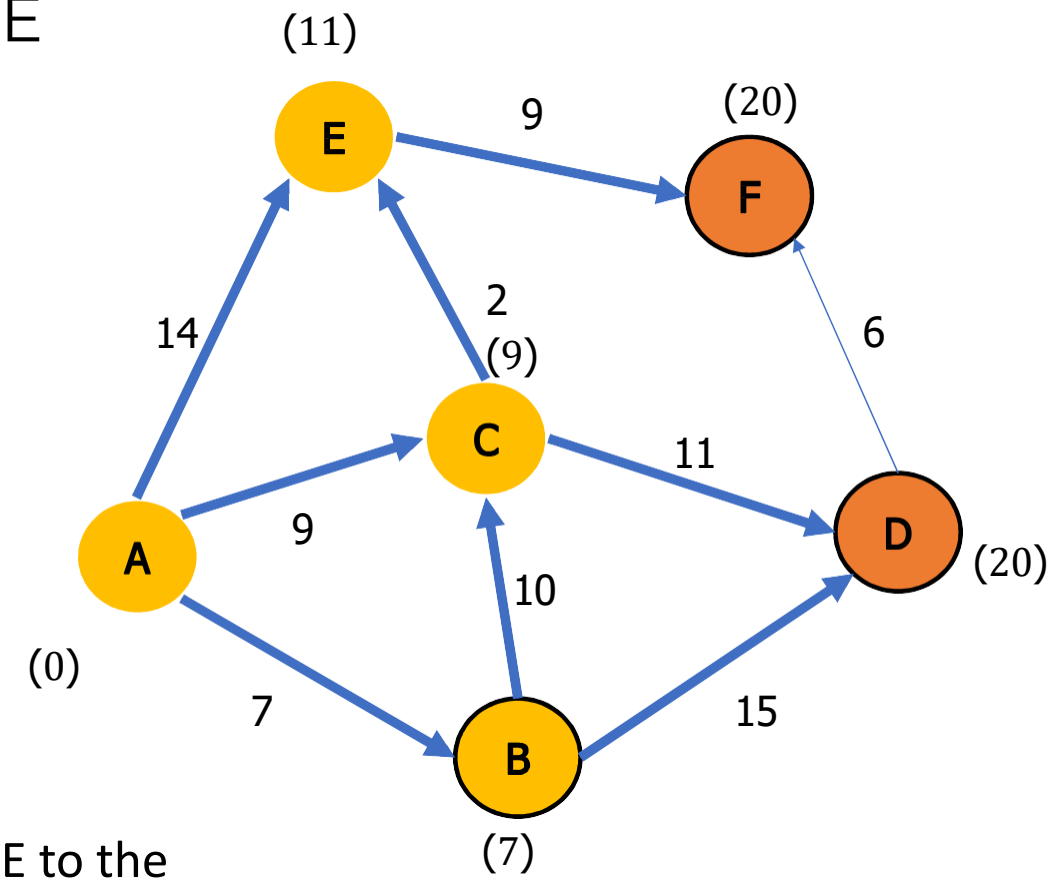


Visited Nodes: [A,B,C] Unvisited Nodes: [D,E,F]

Dijkstra's algorithm

- Step 4: repeat steps 2 and 3 from node E

Node	Shortest Distance	Previous Node
A	0	
B	7	A
C	9	A
D	20	C
E	11	C
F	20	E



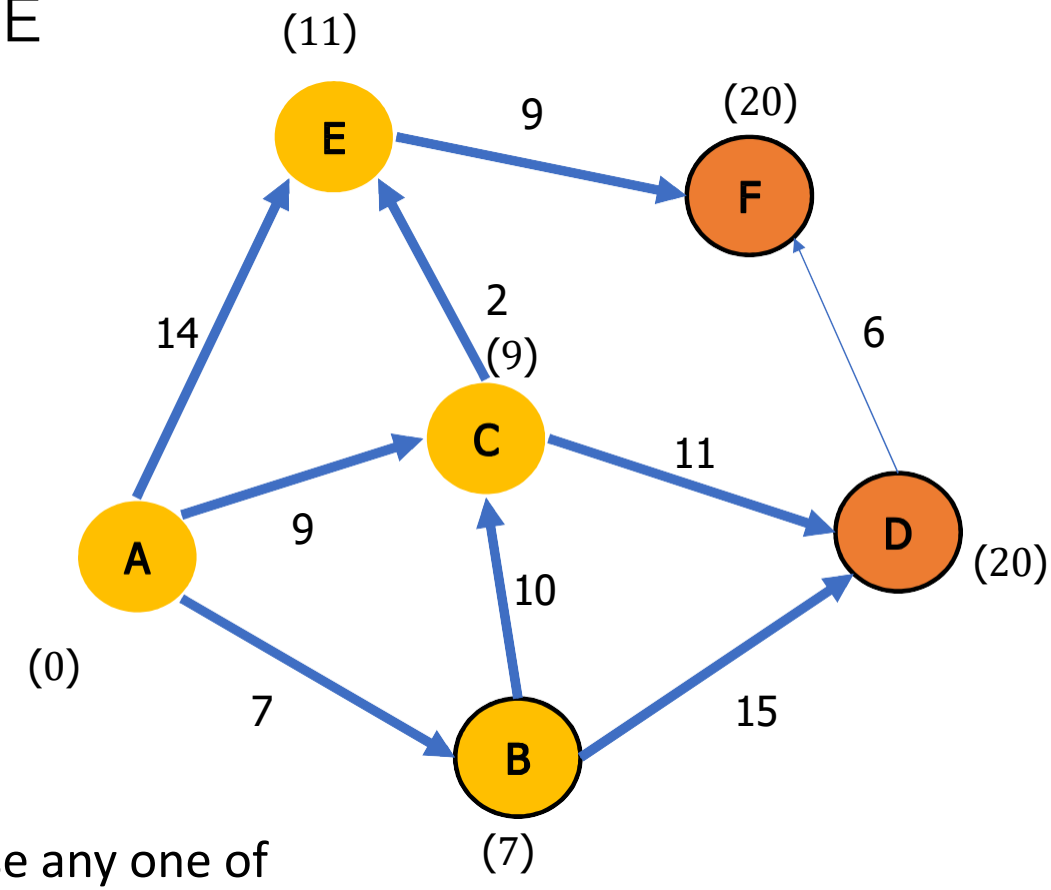
- Once all distances of E's neighbours are updated, we add E to the Visited Nodes list

Visited Nodes: [A,B,C,E] Unvisited Nodes: [E,F]

Dijkstra's algorithm

- Step 4: repeat steps 2 and 3 from node E

Node	Shortest Distance	Previous Node
A	0	
B	7	A
C	9	A
D	20	C
E	11	C
F	20	E



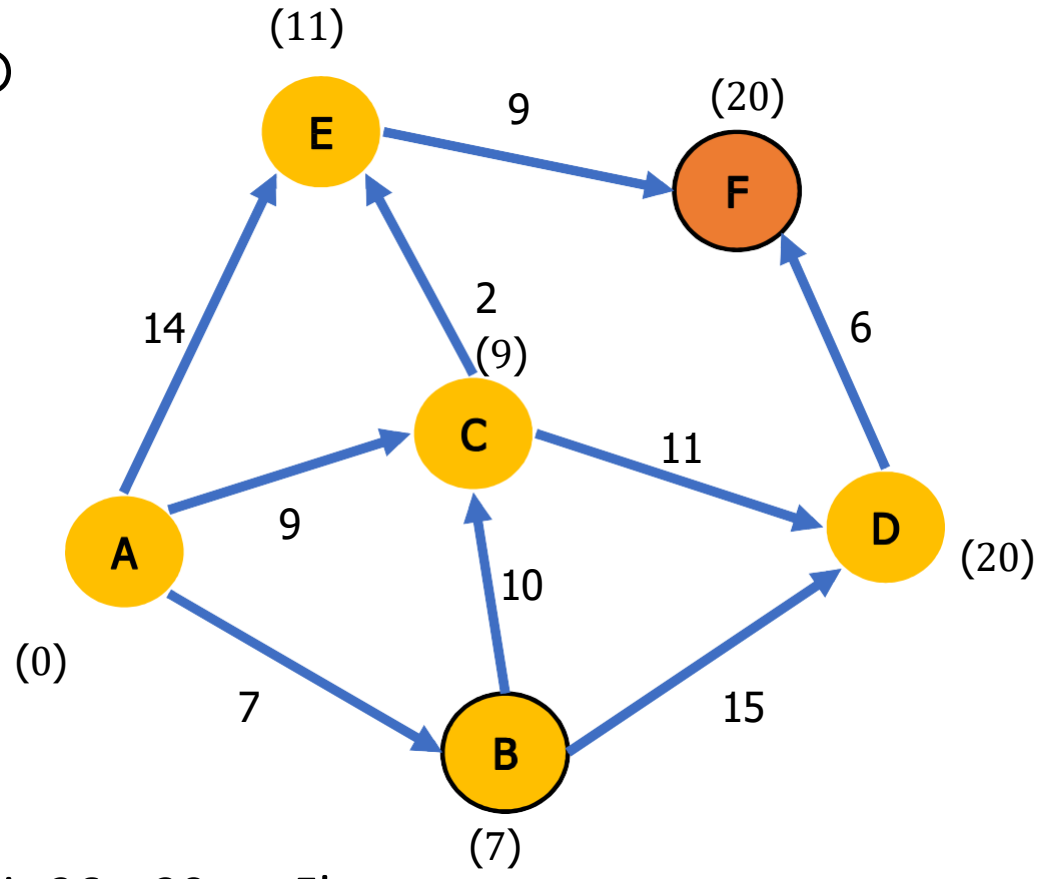
- Since both D and F have the same distance, we can choose any one of the two. Let's choose D as the next node

Visited Nodes: [A,B,C,E] Unvisited Nodes: [E,F]

Dijkstra's algorithm

- Step 4: repeat steps 2 and 3 from node D

Node	Shortest Distance	Previous Node
A	0	
B	7	A
C	9	A
D	20	C
E	11	C
F	20	E



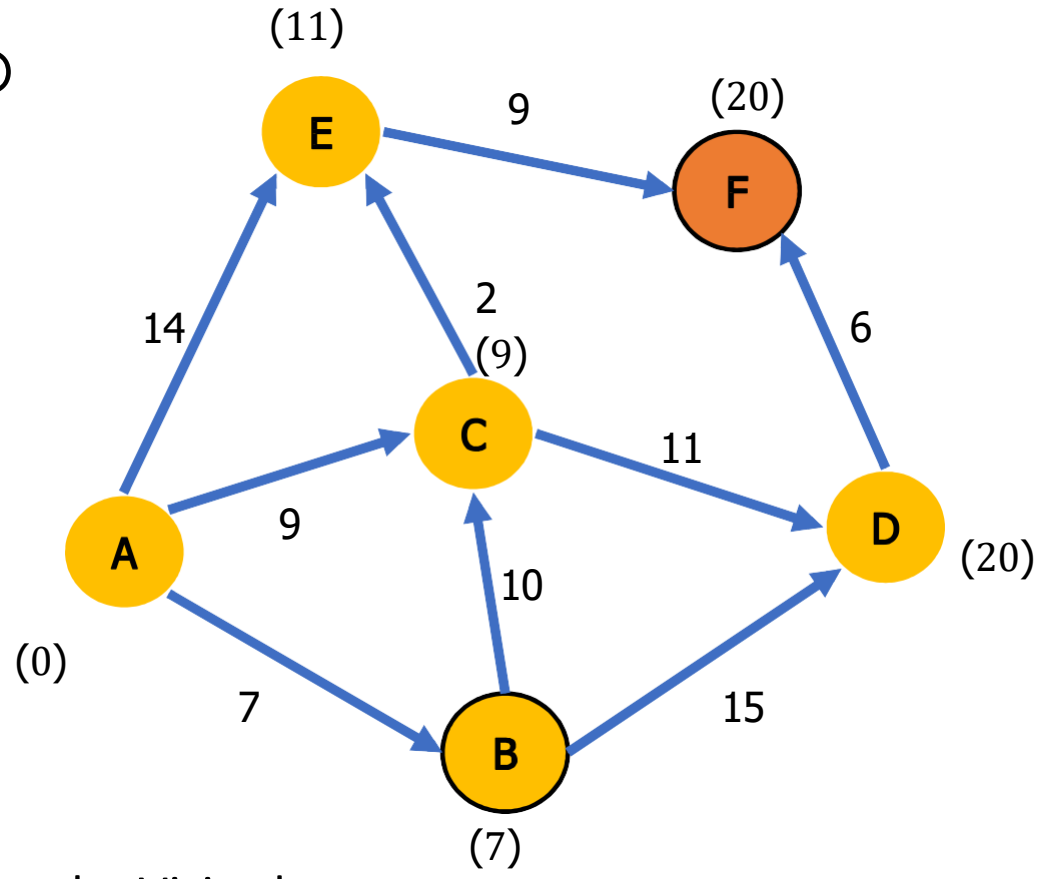
- D has only F as a neighbour and the distance to F through D is 26 > 20, so F's entry in the table won't be updated

Visited Nodes: [A,B,C,E] Unvisited Nodes: [E,F]

Dijkstra's algorithm

- Step 4: repeat steps 2 and 3 from node D

Node	Shortest Distance	Previous Node
A	0	
B	7	A
C	9	A
D	20	C
E	11	C
F	20	E



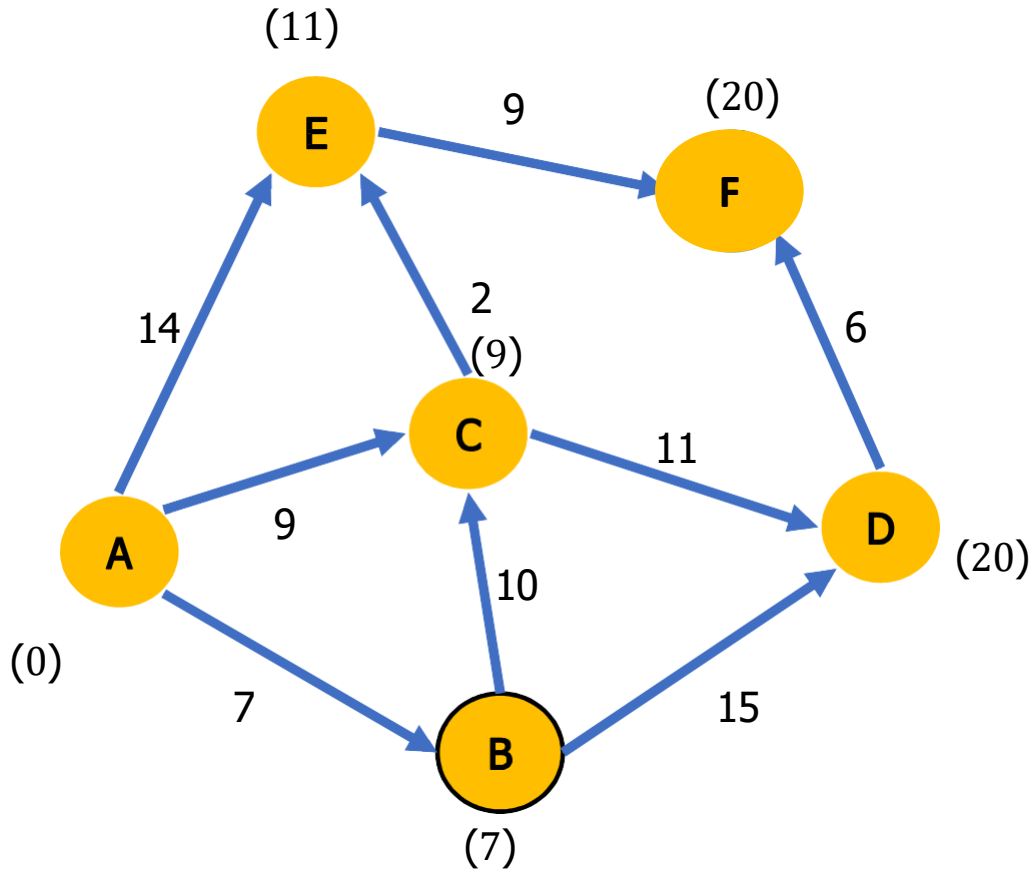
- Once all distances of D's neighbours are updated, we add D to the Visited Nodes list

Visited Nodes: [A,B,C,E,D] Unvisited Nodes: [F]

Dijkstra's algorithm

- Step 4: repeat steps 2 and 3 from node F

Node	Shortest Distance	Previous Node
A	0	
B	7	A
C	9	A
D	20	C
E	11	C
F	20	E

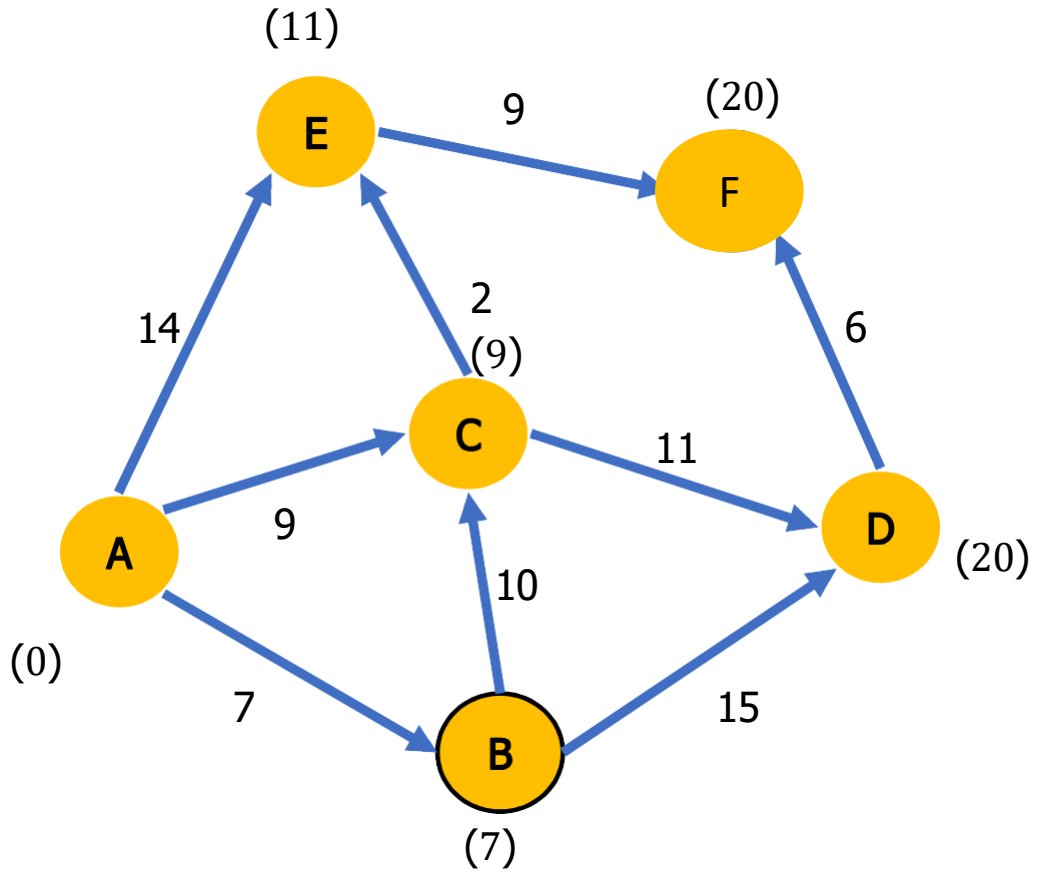


Visited Nodes: [A,B,C,E,D] Unvisited Nodes: [F]

Dijkstra's algorithm

- Step 4: repeat steps 2 and 3 from node F

Node	Shortest Distance	Previous Node
A	0	
B	7	A
C	9	A
D	20	C
E	11	C
F	20	E



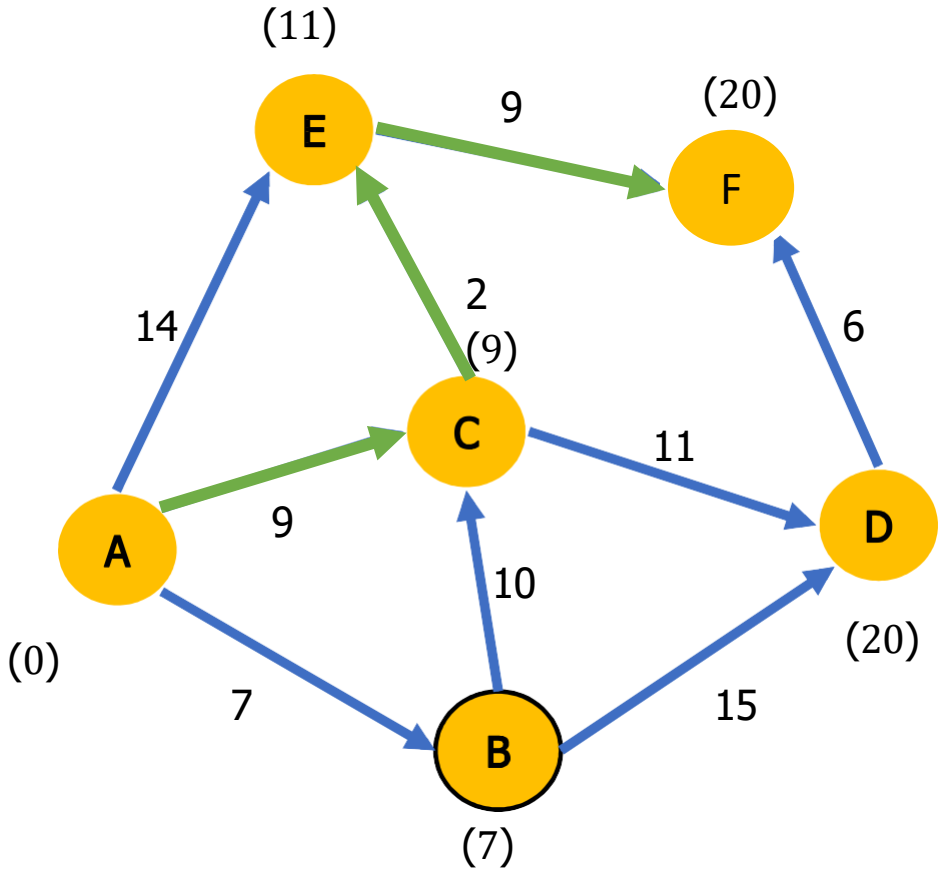
- Since F has no neighbours, we add it to the Visited Nodes list

Visited Nodes: [A,B,C,E,D,F] Unvisited Nodes: []

Dijkstra's algorithm

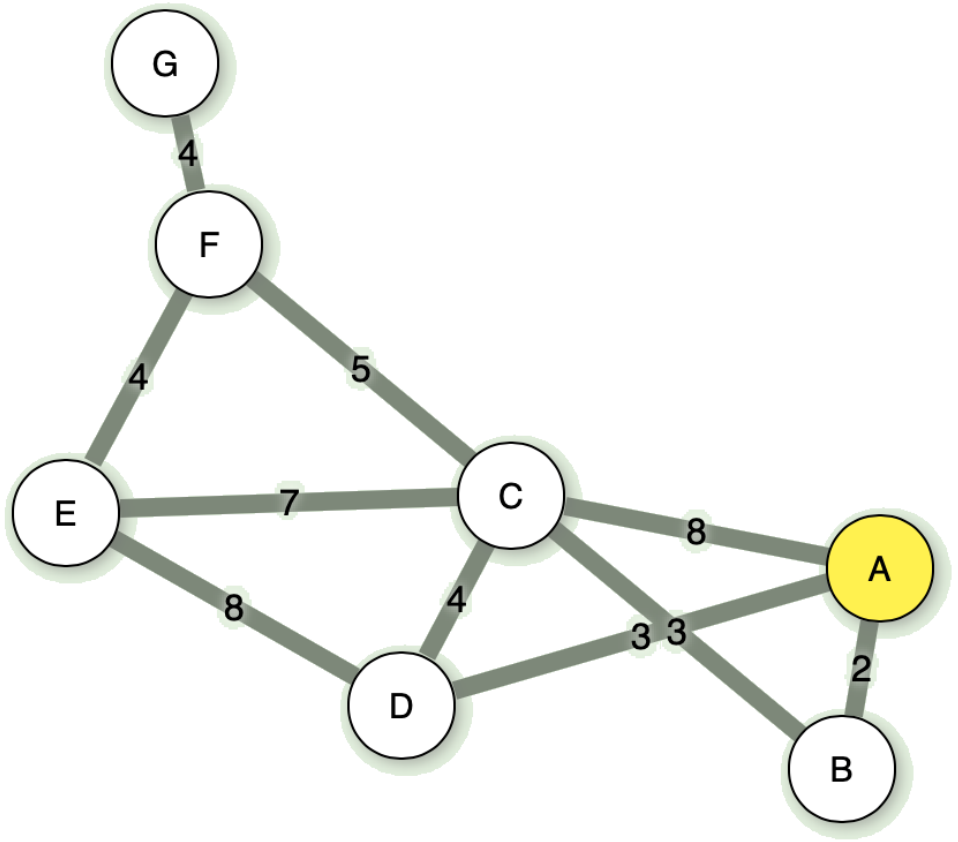
- Therefore, the shortest path from A to F is

  A → C → E → F with a distance of 20



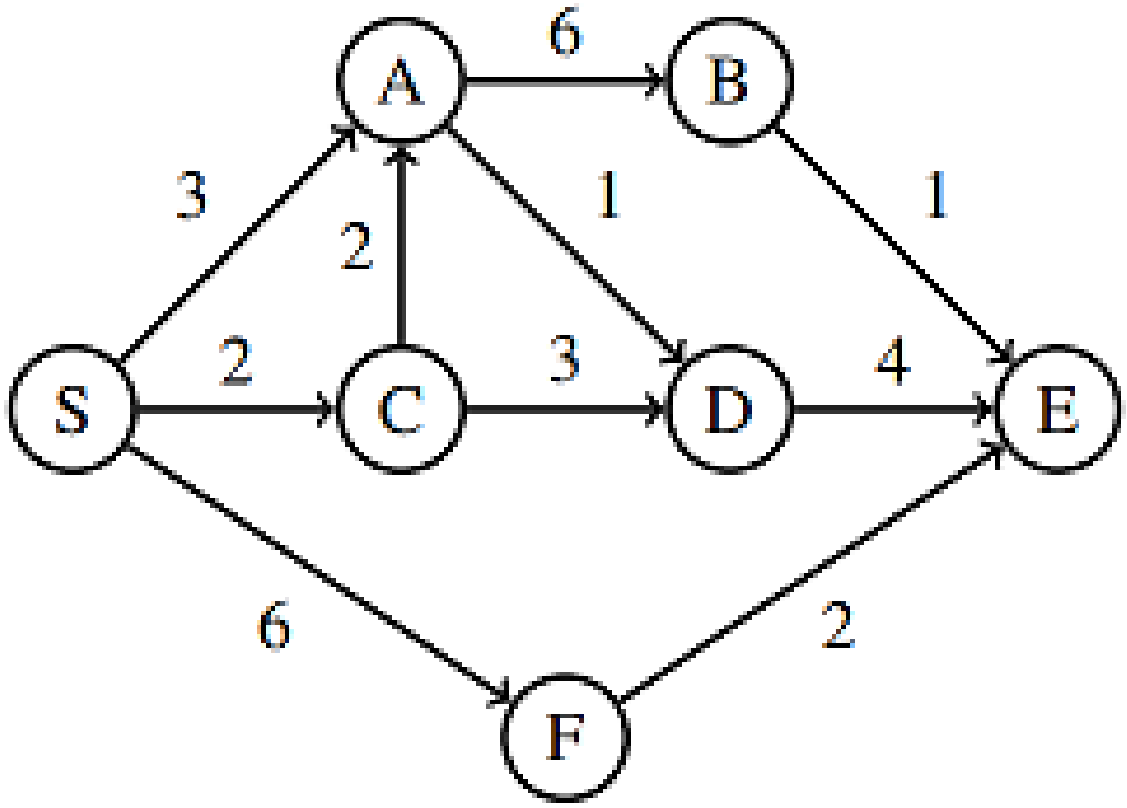
Exercise 1

- Find the shortest path from A to G in the following graph using Dijkstra's algorithm



Exercise 2

- Find the shortest paths from S to all nodes in the following graph using Dijkstra's algorithm



DCU
Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University

THANK YOU